

CHRISTINE DOWLING

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GitHub: github.com/cadowling/project-portfolio | Portfolio: cadowling.github.io

PROFESSIONAL SUMMARY

Analytical and multidisciplinary data scientist with hands-on experience developing predictive models, machine learning algorithms, and end-to-end analytical pipelines across financial, healthcare, and technology applications. Strong foundation in statistical analysis, feature engineering, model development and evaluation, deep learning, and retrieval-based AI with experience working with both structured and unstructured data. Skilled at translating complex technical analysis into actionable insights and collaborating across technical, clinical, and business teams, complemented by experience in product strategy, risk management, finance, and corporate law.

CORE SKILLS

Data Analysis & Modelling: Exploratory Data Analysis, Statistical Analysis, Data Visualization, Feature Engineering, Predictive Modelling, Machine Learning, Deep Learning, Anomaly Detection, Model Tuning, Model Validation & Evaluation

Programming & Data: Python, SQL, MATLAB, PostgreSQL, DuckDB, Tableau

Libraries & Frameworks: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, XGBoost, PyTorch, Tensorflow, Keras, OpenCV

Cloud & Development: AWS (S3, SageMaker, Lambda), Git/GitHub, Jupyter, VSCode

Generative AI: RAG, Embeddings, Vector Search, LangChain, Chroma

Skills: Critical Thinking, Business Analysis, Cross-Functional Collaboration, Data Storytelling, Strategic Planning

EDUCATION

Bachelor of Science, Engineering (With Distinction) — University of Calgary | 2025

Juris Doctor — Queen's University | 2013

Bachelor of Commerce (Risk Management & Finance) (With Distinction) — University of Calgary | 2009

EXPERIENCE

Research Associate (Junior Machine Learning Developer)

University of Calgary (EyeCareX Technologies Ltd.) | Aug 2024 – June 2026

- Developed an end-to-end computer vision pipeline combining deep learning, image processing, and deterministic rules to measure eye movement and identify abnormal patterns.
- Reviewed and audited 100+ patient videos for quality and completeness and built an automated report to track data collection and identify incomplete or unsuitable observations prior to model training and development.
- Engineered and evaluated processing and modelling approaches for patient video data, iterating on algorithms based on observed performance and failure modes.
- Collaborated with clinicians and engineers to validate model outputs against real-world outcomes and support deployment to production.

Product Strategy Advisor & Contract Manager

Willowglen Systems Inc. | May 2022 – Mar 2024

- Conducted market and competitive research and analyzed customer requirements to identify product opportunities and inform prioritization and roadmap decisions.
- Led a bi-weekly meeting with stakeholders from product, marketing, sales and engineering to discuss product development, current blockers, and feature requests; synthesizing quantitative and qualitative findings and presenting recommendations to the executive team on a quarterly basis.
- Developed a structured risk matrix to assess and score contract clauses, applying analytical and risk-based approaches to complex business information.

Corporate Lawyer (Banking Group)

Norton Rose Fulbright | Apr 2017 – Aug 2020

- Advised lenders and borrowers on complex financing transactions, analyzing financial, legal, and contractual issues and delivering concise recommendations to clients and transaction teams.

- Managed large volumes of technical documentation and coordinated with multiple stakeholders to assess transaction risks, resolve issues, and meet time-sensitive closing requirements.

Corporate Lawyer (Financial Services Group)

Borden Ladner Gervais LLP | Jul 2013 – Mar 2017

- Advised corporate clients on credit arrangements, analyzing and negotiating loan and security documentation, and communicating complex financial and legal issues clearly to clients and internal teams.

Insurance Broker

Aon | May 2009 – Aug 2010

- Advised clients on insurance coverage and risk considerations, assessing client requirements and insurance products to develop practical recommendations for corporate risk management programs.

SELECTED DATA SCIENCE & MACHINE LEARNING PROJECTS

Fraud Detection | 2026 (full write-up at cadowling.github.io)

- Conducted exploratory data analysis and built an XGBoost fraud detection model on a highly imbalanced credit card dataset (284K transactions, 0.17% fraud rate), achieving 0.97 ROC-AUC and 0.82 PR-AUC
- Improved precision from 93.6% to 97.3% at equal recall through Bayesian hyperparameter tuning and threshold optimization, cutting false alarms by 60%.
- Applied cost-aware threshold analysis to quantify the trade-off between fraud detection, false alarms, and investigative workload, supporting business-aligned model operating-point selection.

AML Transaction Monitoring & Program Effectiveness | 2026 (full write-up at cadowling.github.io)

- Built an end-to-end AML transaction monitoring system on a 9.5M-transaction synthetic banking dataset, combining deterministic rules with an XGBoost risk-scoring model (0.86 PR-AUC) to identify potentially suspicious activity.
- Conducted exploratory analysis and engineered transaction-level behavioural features, evaluating alternative detection strategies using model-performance and operational-capacity metrics.
- Compared rules-only, machine-learning triaged, and hybrid approaches at matched analyst capacity, increasing laundering detection from 6.9% to 77.6%, demonstrating the impact of model design and operating thresholds on detection and alert workload.
- Built Tableau dashboards translating alert volume, analyst workload, and typology-coverage metrics into program-effectiveness insights for technical and non-technical stakeholders.

AI-Help Desk Assistant Using RAG | 2026 (full write-up at cadowling.github.io)

- Built an end-to-end Retrieval-Augmented Generation (RAG) system using Python, LangChain, OpenAI embeddings, Chroma vector search, and an LLM to answer questions using organization-specific documentation.
- Designed the retrieval pipeline including document loading, semantic chunking, embedding generation, persistent vector storage, similarity-based retrieval, relevance thresholds, and grounded response generation.
- Implemented conversational memory while maintaining strict grounding rules and safe fallback behaviour when information was not supported by the retrieved context.

A/B Testing & Statistical Analysis | 2026 (full write-up at cadowling.github.io)

- Built a Python-based A/B testing analysis to assess whether promotional mailer type influenced customer signup rates, using Pandas for data preparation and SciPy for statistical testing.
- Applied Chi-Square Test for Independence to formulate and evaluate hypotheses, calculating test statistics, p-values, expected frequencies, and critical values at a 0.05 significance level.
- Translated statistical findings into actionable business insights, evaluating whether observed differences were sufficiently supported by the data to inform campaign and ROI decisions.